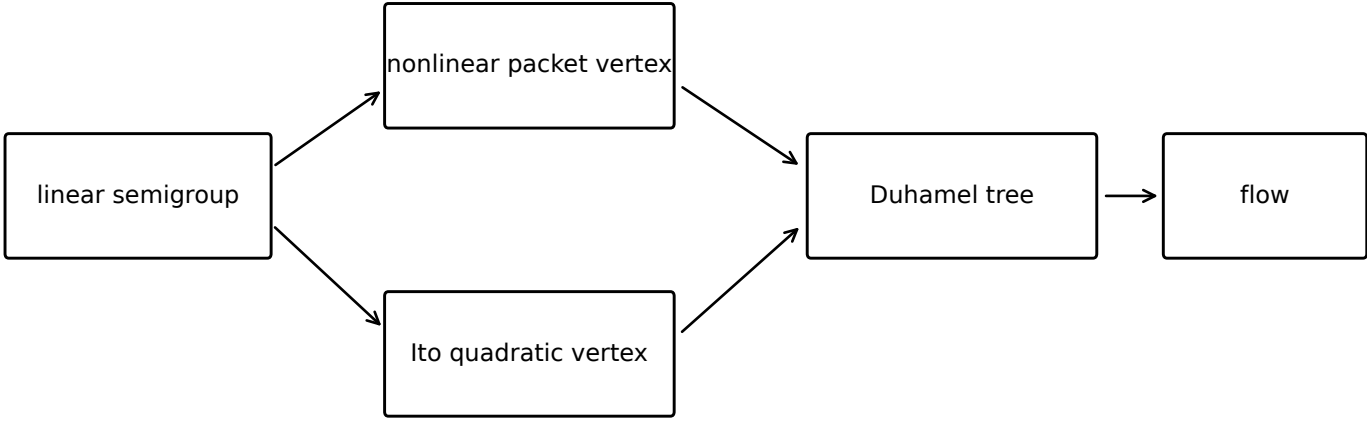Resolved dynamics uses linear edges, nonlinear hyperedges, and stochastic contractions